上海科技大学
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CS240 Algorithm Design and Analysis

Lecture 26

Approximation Algorithms

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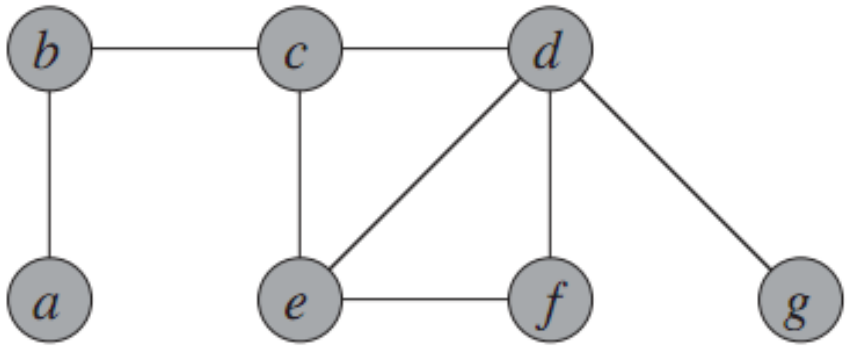
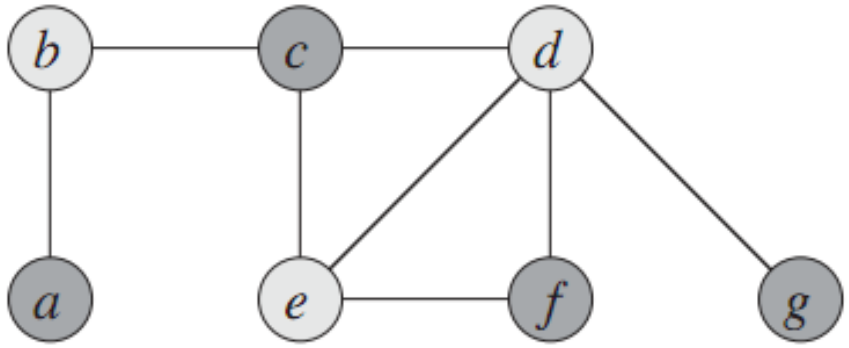
Vertex Cover

Vertex Cover

- **Input** A graph with vertices V and edges E.
- **Output** A subset V’ of the vertices, so that every edge in E touches some vertex in V’.
- **Goal** Make |V’| as small as possible.



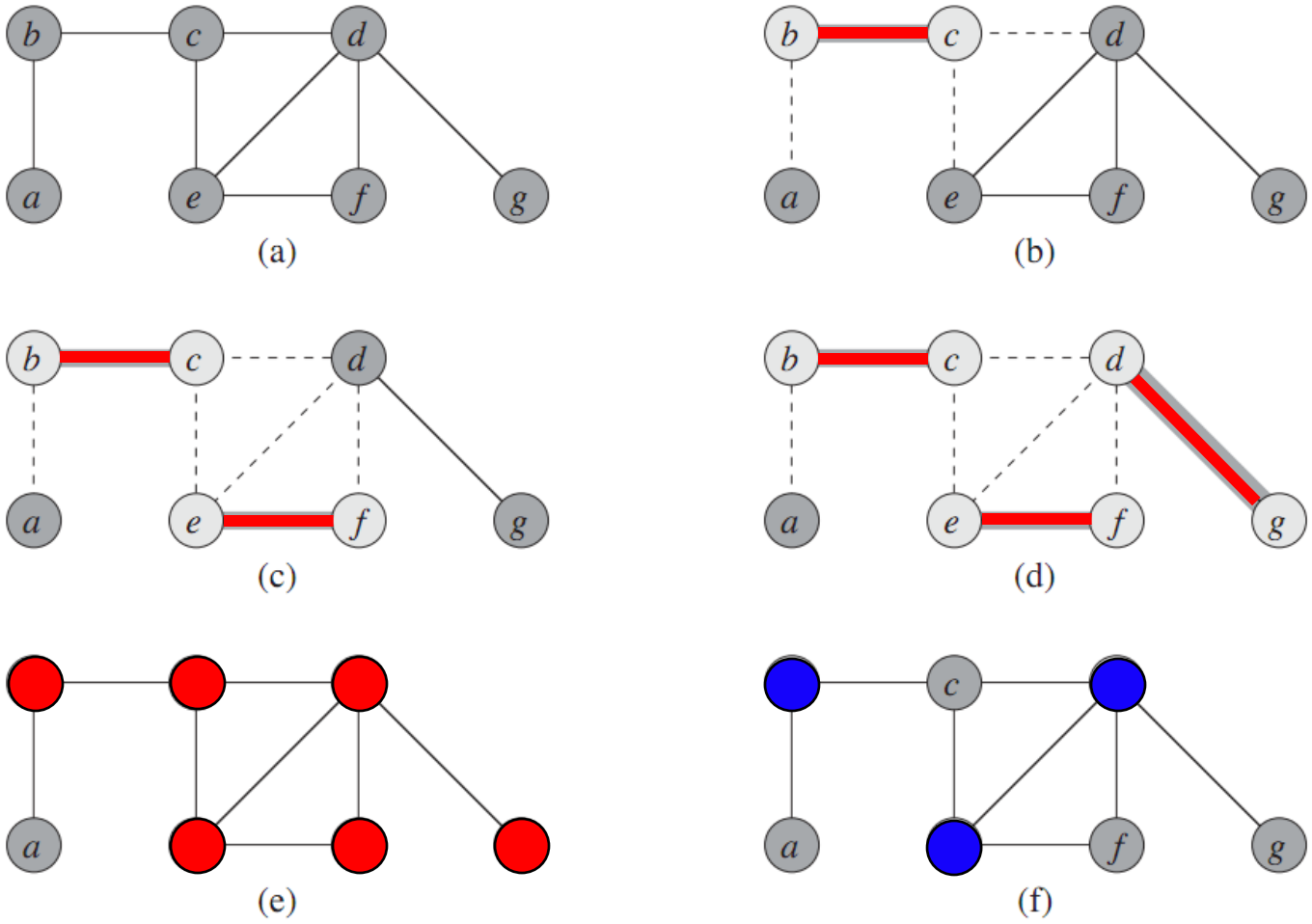


- Finding the minimum vertex cover is NP-complete.
- We’ll see a simple 2 approximation for this problem.

A Vertex Cover Algorithm

- Initially, let D be all the edges in the graph, and C be the empty set.
  - C is our eventual vertex cover.
- Repeat as long as there are edge left in D.
  - Take any edge (u,v) in D.
  - Add {u,v} to C.
  - Remove all the edges adjacent to u or v from D.
- Output C as the vertex cover.

Example



Algorithm's vertex cover

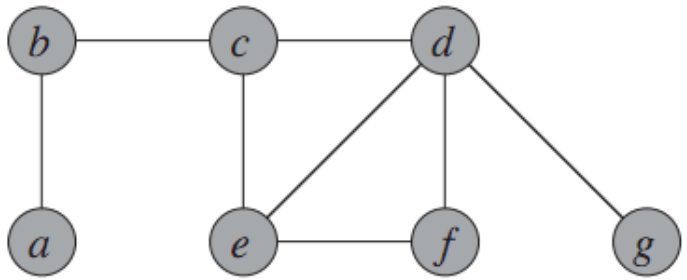
Optimal vertex cover

Proof of Correctness

- The output is certainly a vertex cover.
  - In each iteration, we only take out edges that get covered.
  - We keep adding vertices till all edges are covered.
- Now, we show it's a 2 approximation.
- Let C* be an optimal vertex cover.
- Let A be the set of edges the algorithm picked.

Proof of Correctness

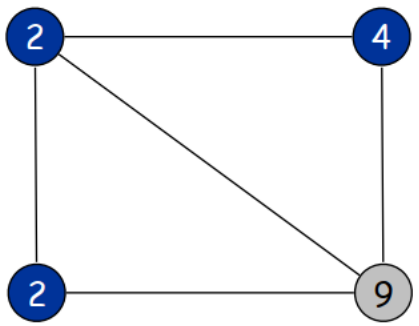
- None of the edges in A touch each other.
  - Each time we pick an edge, we remove all adjacent edges.
- So each vertex in C* covers at most one edge in A.
  - The edges covered by a vertex all touch each other.
- Every edge in A is covered by a vertex in C*.
  - Because C* is a vertex cover.
- So $|C^*| \geq |A|$.
- The number of vertices the algorithm uses is $2|A|$.
  - If algorithm picks edge (u,v), it uses {u,v} in the cover.
- So (# vertices algorithm uses) / (# vertices in opt cover) = $2|A| / |C^*| \leq 2|A| / |A| = 2$.



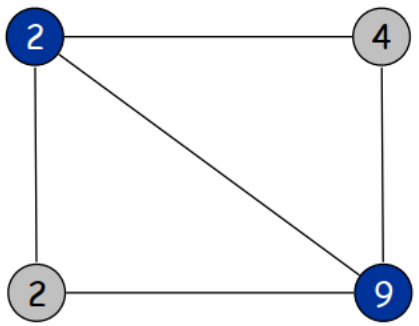
The Pricing Method: Vertex Cover

Weighted Vertex Cover

Weighted vertex cover. Given a graph G with vertex weights, find a vertex cover of minimum weight. It's a special case of the set cover problem, so the H(d*) approximation ratio can be achieved by the greedy algorithm, where d* = max degree



weight = 2 + 2 + 4 = 8



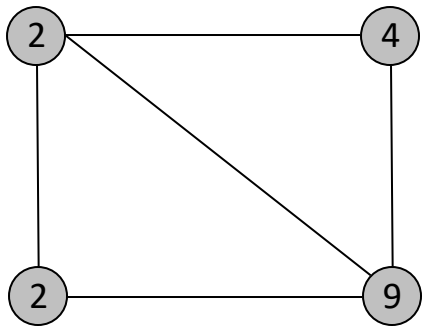
weight = 2 + 9 = 11

Weighted Vertex Cover

Pricing method. Each edge must be covered by some vertex i. Edge e pays price $p_e \geq 0$ to use vertex i.

Fairness. Edges incident to vertex i should pay $\leq w_i$ in total.

$$for\ each\ vertex\ i: \sum_{e=(i,j)} p_e \leq w_i$$



Claim. For any vertex cover S and any fair prices p_e: $\sum_e p_e \leq w(S)$.

Proof.

$$\sum_{e \in E} p_e \leq \sum_{i \in S} \sum_{e=(i,j)} p_e \leq \sum_{i \in S} w_i = w(S)$$

each edge e covered by at least one node in S

sum fairness inequalities for each node in S

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Pricing Method

Pricing method. Set prices and find vertex cover simultaneously.

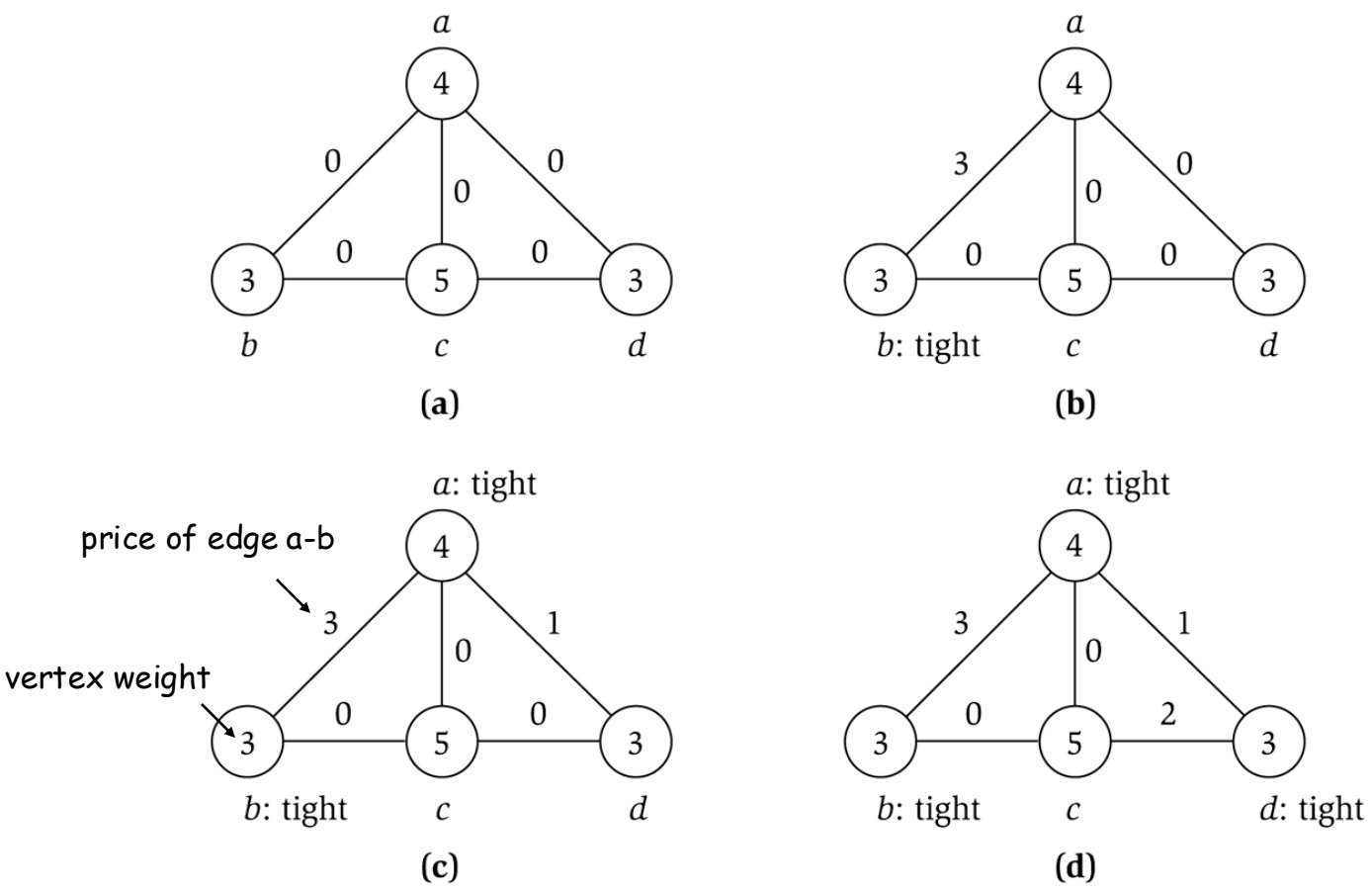
```
Weighted-Vertex-Cover-Approx(G, w) {
   foreach e in E
      pₑ = 0

   while (∃ edge i-j such that neither i nor j are tight)
      select such an edge e
      increase pₑ without violating fairness
   }

   S ← set of all tight nodes
   return S
}
```

$$\sum_{e=(i,j)} p_e = w_i$$

Pricing Method: Example



Pricing Method: Analysis

Theorem. Pricing method is a 2-approximation.

Pf.

- Algorithm terminates since at least one new node becomes tight after each iteration of while loop.

- Let S = set of all tight nodes upon termination of algorithm.
- S is a vertex cover: if some edge i-j is uncovered, then neither i nor j is tight. But then while loop would not terminate.

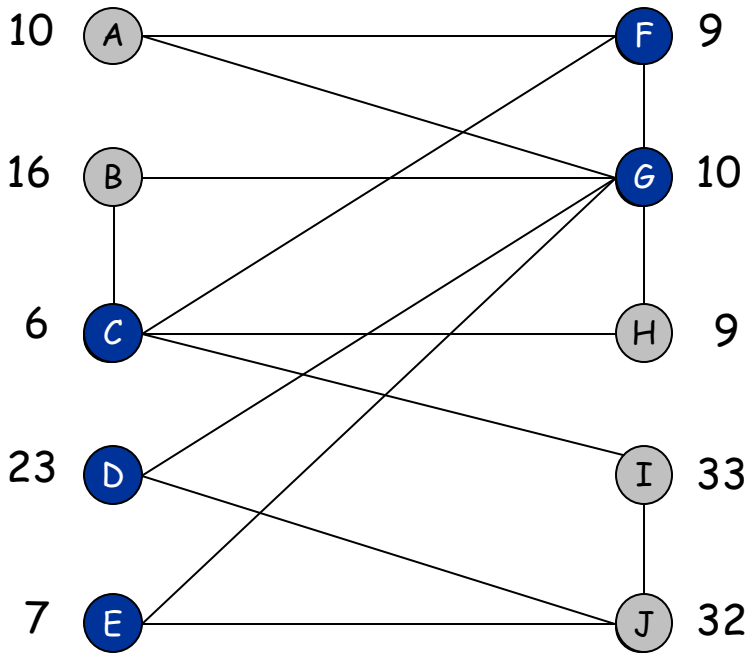
- Let S* be optimal vertex cover. We show $w(S) \leq 2w(S^*)$.

$$w(S) = \sum_{i \in S} w_i \underset{\text{all nodes in S are tight}}{=} \sum_{i \in S} \sum_{e=(i,j)} p_e \underset{\substack{S \subseteq V, \\ \text{prices} \geq 0}}{\leq} \sum_{i \in V} \sum_{e=(i,j)} p_e \underset{\text{each edge counted twice}}{=} 2 \sum_{e \in E} p_e \underset{\text{fairness lemma}}{\leq} 2w(S^*). \quad \blacksquare$$

LP Rounding: Vertex Cover

Weighted Vertex Cover

Weighted vertex cover. Given an undirected graph G = (V, E) with vertex weights $w_i \geq 0$, find a minimum weight subset of nodes S such that every edge is incident to at least one vertex in S.



total weight = 55

Weighted Vertex Cover: Integer Linear Programming Formulation

Weighted vertex cover. Given an undirected graph G = (V, E) with vertex weights $w_i \geq 0$, find a minimum weight subset of nodes S such that every edge is incident to at least one vertex in S.

Integer programming formulation.

- Model inclusion of each vertex i using a 0/1 variable x_i.

$$x_i = \begin{cases} 0 & \text{if vertex } i \text{ is not in vertex cover} \\ 1 & \text{if vertex } i \text{ is in vertex cover} \end{cases}$$

- Objective function: minimize $\sum_i w_i x_i$.
- Must take either i or j: $x_i + x_j \geq 1$.

$$\begin{array}{llll} (ILP) \ \min & \sum_{i \in V} w_i x_i & & \\ \text{s.t.} & x_i + x_j & \geq 1 & (i,j) \in E \\ & x_i & \in \{0,1\} & i \in V \end{array}$$

Integer Programming

INTEGER-PROGRAMMING. Given integers a_{ij} and b_i, find integers x_j that satisfy:

$$\begin{array}{lll} \min & c^t x & \\ \text{s.t.} & Ax & \geq & b \\ & x & \geq & 0 \\ & x \text{ integral} & & \end{array}$$

$$\begin{array}{llll} \sum_{j=1}^{n} a_{ij} x_j & \geq & b_i & 1 \leq i \leq m \\ x_j & \geq & 0 & 1 \leq j \leq n \\ x_j & & \text{integral} & 1 \leq j \leq n \end{array}$$

Observation. Vertex cover formulation proves that integer programming is NP-hard search problem.

even if all coefficients are 0/1 and
at most two variables per inequality

Integer Programming

Linear programming. Max/min linear objective function subject to linear inequalities.

- Input: integers c_j, b_i, a_{ij} .
- Output: **real numbers** x_j.

$$\begin{array}{llll} \text{(LP)} & \min & c^t x & \\ & \text{s. t.} & Ax & \geq & b \\ & & x & \geq & 0 \end{array}$$

$$\begin{array}{llll} \text{(LP)} & \min & \sum_{j=1}^{n} c_j x_j & & \\ & \text{s. t.} & \sum_{j=1}^{n} a_{ij} x_j & \geq & b_i \quad 1 \leq i \leq m \\ & & x_j & \geq & 0 \quad 1 \leq j \leq n \end{array}$$

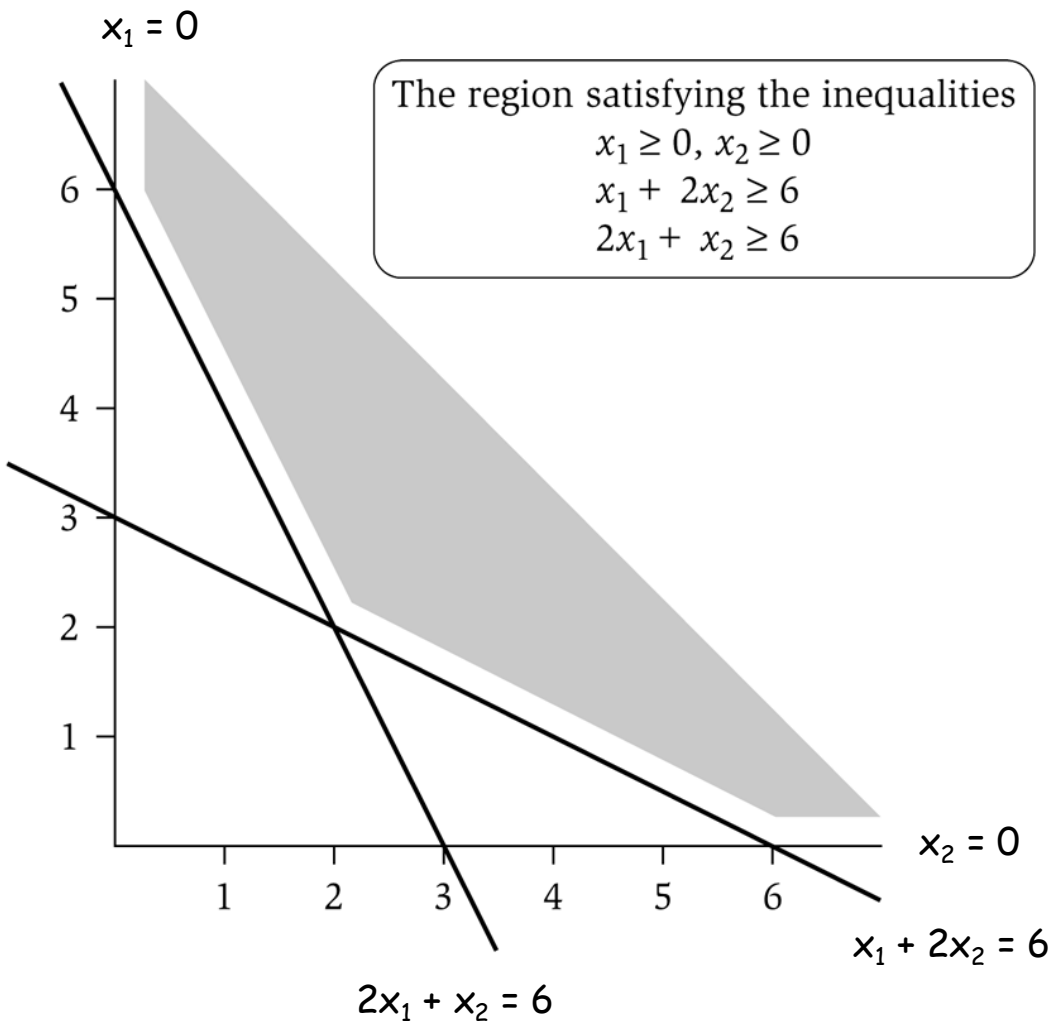
Linear. No x^2, xy, arccos(x), x(1-x), etc.

Simplex algorithm. [Dantzig 1947] Can solve LP in practice.

Ellipsoid algorithm. [Khachian 1979] Can solve LP in poly-time.

LP Feasible Region

LP geometry in 2D.



Weighted Vertex Cover: LP Relaxation

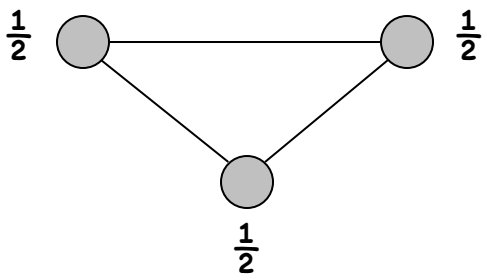
Weighted vertex cover. Linear programming formulation.

$$
\begin{array}{llll}
(LP) \ \min & \sum_{i \in V} w_i x_i & & \\
\text{s.t.} & x_i + x_j & \geq 1 & (i,j) \in E \\
& x_i & \geq 0 & i \in V
\end{array}
$$

Observation. Optimal value of (LP) is $\leq$ optimal value of (ILP).

Pf. LP has fewer constraints.

Note. LP is not equivalent to vertex cover.



Q. How can solving LP help us find a small vertex cover?

A. Solve LP and **round** fractional values.

Weighted Vertex Cover

Theorem. If x^* is optimal solution to (LP), then $S = \{i \in V : x^*_i \geq ½\}$ is a vertex cover whose weight is at most twice the min possible weight.

Pf. [S is a vertex cover]

- Consider an edge $(i, j) \in E$.
- Since $x^*_i + x^*_j \geq 1$, either $x^*_i \geq ½$ or $x^*_j \geq ½$ $\Rightarrow$ (i, j) covered.

Pf. [S has desired cost]

- Let S^* be optimal vertex cover. Then

$$\sum_{i \in S^*} w_i \geq \sum_{i \in S} w_i x_i^* \geq \frac{1}{2} \sum_{i \in S} w_i$$

LP is a relaxation

$x_i^* \geq \frac{1}{2}$